1. Clinical Question & Core Hypothesis

Clinical Question

Pretreatment tumors with high lymphocytic infiltration often show variable cisplatin sensitivity — why does immune presence (TILs) not uniformly translate to better chemotherapy response? 2) Tier 1 paths used: ERCC1-mediated DNA repair (Core Path 1) as the tumor-intrinsic mechanism that can neutralize cisplatin-induced immunogenic DNA damage, and the urothelial differentiation program (KRT20 signature, Core Path 3) as a modifier of both repair programs and immune-visibility. I built on these because they provide a clear mechanistic axis (repair capacity × differentiation state) that can blunt or permit TIL-mediated tumor clearance after platinum. 3) Tier 2 evidence supporting the claim: ERCC2 and TP53 pathways (Tier 2) reinforce the DNA-repair/checkpoint axis and can refine prediction of repair competence; CD8A/GZMB (Tier 2) support the immune-effector side and will be used to validate that TIL imaging corresponds to cytotoxic potential. 4) Endpoint choice rationale: I selected therapy_response (cisplatin response proxy — e.g., pathologic downstaging/pCR or radiographic response) because the seed KG and proposed clinical application focus on predicting chemotherapy sensitivity from pre-treatment tissue; therapy response is the most directly actionable endpoint for MGX in the neoadjuvant/trial context. 5) Deliberate exclusions: I excluded CD8A and GATA3 from the core anchor pair to satisfy the 1–2 anchor-gene constraint and because TIL density is captured as the imaging variable; CD8A is therefore placed among secondary_modifiers to be used for molecular validation. ERCC2, TP53, and FGFR3 are also delegated to secondary_modifiers to keep the hypothesis sharp and testable while preserving strong mechanistic candidates for follow-up?

Working Hypothesis

In pretreatment urothelial carcinomas treated with cisplatin-based regimens, the predictive value of tumor-infiltrating lymphocytes (TILs) for therapy response is contingent on the tumor's intrinsic DNA repair capacity and its differentiation state. Specifically, high TIL density predicts favorable cisplatin response predominantly in tumors with low ERCC1-mediated nucleotide-excision repair and/or a non-luminal differentiation program (reflected by a lower KRT20 signature). Conversely, tumors with high ERCC1 expression and strong luminal differentiation (high KRT20) and with intact ERCC2 and TP53 signaling will exhibit resistance to cisplatin despite abundant TILs, because efficient repair and a differentiation-driven immune-excluded microenvironment limit immunogenic cell death and TIL effectiveness. To test this, we propose an MGX (molecular-genomic-exposure) rule that integrates: (i) imaging-based TIL density as a measure of immune presence, (ii) ERCC1 expression as a proxy for NER-driven repair capacity, (iii) KRT20 signature as a differentiation-state modifier, and (iv) secondary modifiers including ERCC2 and TP53 status (repair/checkpoint integrity), FGFR3 signaling as an additional differentiation-associated pathway, and immune effector markers (CD8A, GZMB) for validation of cytotoxic potential. The anchor pair remains ERCC1 and KRT20, while CD8A and GZMB serve as molecular validation of effector activity, and CD8A is kept in secondary modifiers rather than anchors to preserve the 1–2 anchor constraint. The hypothesis yields testable predictions: (a) high TIL density combined with low ERCC1 and/or low KRT20 will associate with higher rates of pathologic complete response (pCR) or major radiographic response to neoadjuvant cisplatin; (b) high ERCC1 expression and high KRT20 (with intact ERCC2/TP53) will predict poor response despite high TILs; (c) ERCC2 or TP53 loss-of-function will shift the expected outcome toward greater chemosensitivity in contexts of high TIL density and/or altered differentiation. Feasible testing approaches include collecting pre-treatment biopsies with matched clinical outcomes, performing quantitative digital pathology for TIL density, assessing ERCC1/ERCC2 expression or function (RNA/protein), profiling KRT20-associated differentiation, sequencing for ERCC2/TP53/FGFR3 status, and measuring immune-effector markers (CD8A, GZMB). The MGX score will be evaluated for predictive performance against standard clinical predictors, with validation in independent cohorts. This framework provides mechanistic plausibility (DNA repair capacity × differentiation state modulating immune-driven clearance) and a concrete path to clinical stratification for neoadjuvant therapy.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	ERCC1, KRT20
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ERCC1: ERCC1-mediated DNA repair (Core Path 1) as the tumor-intrinsic mechanism that can neutralize cisplatin-induced immunogenic DNA damage, and the urothelial differentiation program (KRT20 signature, Core Path 3) as a modifier of both repair programs and immune-visibility.

KRT20: KRT20 signature, Core Path 3) as a modifier of both repair programs and immune-visibility.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CD8A	Supporting modifier from knowledge graph
Secondary	GATA3	Supporting modifier from knowledge graph
Secondary	ERCC2	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	FGFR3	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: that can be validated in retrospective cohorts and prospectively tested in trials, with clear endpoints such as pCR or major radiographic response. This strategy remains focused, testable, and grounded in established mechanistic links between DNA repair, differentiation, and anti-tumor immunity.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CD8A: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- GATA3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.